

Practice Quiz 06D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 06

Prove uniqueness of a complement within a fixed set.

1. Let U be a set and let $A \subseteq U$. The set $U \setminus A$ satisfies the following two conditions on B :

$$A \cap B = \emptyset \quad \text{and} \quad A \cup B = U.$$

Prove that no other subset B of U satisfies both conditions.

Solution Suppose $C \subseteq U$ satisfies $A \cap C = \emptyset$ and $A \cup C = U$. If $x \in C$, then $x \in U$ and $x \notin A$, so $x \in U \setminus A$. Conversely, if $x \in U \setminus A$, then $x \in U$ and $x \notin A$. Since $U = A \cup C$, it follows that $x \in C$. Therefore $C = U \setminus A$, proving uniqueness.

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